

University of Regina

Department of Biology

Biology 835AQ -Data Science in Ecology

August, 21- 2026

Professor: Bruno Eleres Soares

Student: Rodrigo Martin de Oliveira

SN: 200537496

The development of a function in R language to process diet analysis

Introduction

Diet studies are a crucial point in ecology; it is through diet studies that we can understand the trophic position of an organism and how it relates to the environment (Sabo & Gerber, 2021). Usually, in diet studies, researchers use several types of mechanisms to understand an organism's diet. The Importance Alimentary index (IA_i; Kawakami & Vazzoler, 1980) is a tool widely utilized that explains the importance of a food resource on an organism's diet. It takes into consideration the percentage of volume (V%) of a food item and the occurrence (FO%) of it within the fish population. For example, a food item that has a high IA_i is a particularly important alimentary food item. As the index is a correlation between how much a given food item occupies the stomach and how often it happens on the stomach of the population, it brings us to the Alimentary index.

The Important Alimentary index assumes that there is a relation between the products of the frequency of occurrence percentage and the volume percentage of each item, that can be used to estimate their diet importance by adding them on a cartesian plane, where X is FO% and Y is V%, we can have an area and this area will be the IA_i Σ_i is summatory of FO% times VO% of all stomachs of the population (Kawakami & Vazzoler, 1980), like observed on the following expression:

$$IA_i = (F_i \times V_i) / \Sigma_{i=1}^n (F_i \times V_i)$$

where:

- IA_i = food index of item *i*
- F_i = frequency of occurrence (%) of item *i*
- V_i = volume (%) of item *i*
- n = total number of food items analyzed

So, the Alimentary index allows us to correlate the occurrence and the volume and understand better the importance of each food item on a population.

Cleaning protocols

The fast development of search tools to search the internet for data has raised what we call a big data, and with it comes the data disaster (Guo et al., 2023). The data cleaning process is the 'savior' that turns the mass of data into something usable. An untrained scientist can save data in a messy way, potentially making the set unusable. It is a very common issue to have someone spending a lot of time on manual data cleaning and running into the chance of modifying the data in a way that does not reflect reality. A Data-cleaning script is a very important tool to retain the original raw data and return the cleaned version while preserving the data history. For modern reproducible science, an automated protocol tool is crucial (Diederich et al., 2022). It is also important for this automated protocol to be reproducible, so an automated protocol could be beneficial to scientists.

Nowadays, there is no automated protocol to calculate the IAI, which opens the door to problems such as the struggle to deal with large datasets and how easy it is to make mistakes when dealing with a spreadsheet document. To avoid these problems, a tool that could automatically calculate the IAI would bring an advantage, speeding the process and allowing the scientist to go to the next step. So, a clear pipeline of the process could make the whole data analysis clear and reproducible.

The philosophy of the book R Packages, written by Wickham & Bryan and last released in 2025, states that "Anything that can be automated should be automated." And it continues, addressing the need for functions that do more in less time, because it will perform tasks through a trackable protocol, reinforcing good science practices. But as we stand, there is no R package that performs an IAI analysis. So, a clear pipeline that can perform a reproducible cleaning protocol and perform the IAI analysis is needed.

Results

Seven functions were developed:

- validate_diet_data
- clean_diet_data
- run_all_observed_scenarios
- fo_summary
- volume_summary
- iai_summary
- diet_indices_summary

In every function, there is a step for "data frame" verification to prevent consequential errors.

This project was made using the following libraries:

- tidyverse

- janitor

The workflow starts with a validation function that screens the raw database for inconsistencies and data quality and returns a report, followed by a cleaning process that attempts to solve all common problems, such as non-standardization of character encoding. I am choosing UTF-8 as the text encoding for its simplicity and compatibility, e.g. text typed on an IOS environment. At this point, the validation step will run again to assess if the problems were reduced or fixed. If the validation step is cleared, then we pass to the analysis step.

For the data analysis function, it follows the sequence of:

- It identifies every observed combination of the selected metadata variables.
- It filters the dataset for one combination at a time.
- It sends that filtered subset to `fo_summary()`.
- It sends the same subset to `volume_summary()`.
- It sends the same subset to `iai_summary()`.
- It sends the same subset to `diet_indices_summary()`.
- It stores all outputs under the corresponding scenario name.

To use the script clone the [rod_BIOI835AQ](#) repository from [Soares Quantitative Ecology Lab](#), install (if not already installed) tidyverse and janitor, refer to the functions script with `"source("01_script/functions.R")`, load the raw dataset, run the validation, cleaning, revalidation, and run the analysis of the whole data set, if desired, for a specific grouping run `all_scenarios` and select how do you want to group under `combo_vars`, after this combine the results into one database and export if needed using the `write.csv` tool.

Here is a diagram of how the function `all_observed_scenarios` is structured:

Clean_data

|

`run_all_observed_scenarios()`

|

+ filtered scenario 1 - `fo_summary()`

|

- `volume_summary()`

|

- `iai_summary()`

|

- `diet_indices_summary()`

|

```

+ filtered scenario 2 - fo_summary()
|
|           - volume_summary()
|
|           - iai_summary()
|
|           - diet_indices_summary()
|
+ filtered scenario 3 - same analysis functions...

```

Discussion

It is difficult to produce a real-world comparison of an analysis made without the proposed package, but I foresee being able to help scientists spend less time writing and troubleshooting code and allowing them more time to spend interpreting their data. As a scientist myself, I understand the frustration of not being able to accomplish the results you expected. By simplifying the workflow and keeping it clear, trackable, and reproducible, it can help make diet data analysis more accessible to students and researchers.

What could be made better is to implement more index calculations, allowing scientists to use not just for calculating the percentage of volume, occurrence, and importance alimentary index.

Conclusion

This work is successful in outputting results of FO%, V% and IAI%, with validation and cleaning procedures and processing a database from fish gut analysis.

More real-world tests are still needed; this work used biased databases and a randomly generated database.

Bibliography

Diederich, K., Schmitt, K., Schwedhelm, P., Bert, B., & Heint, C. (2022). A guide to open science practices for animal research. **PLoS Biology*, 20*(9), e3001810.

<https://doi.org/10.1371/journal.pbio.3001810>

Firke, S. (2024). **janitor: Simple tools for examining and cleaning dirty data** [R package version 2.2.1]. <https://CRAN.R-project.org/package=janitor>

Guo, M., Wang, Y., Yang, Q., Li, R., Zhao, Y., Li, C., Zhu, M., Cui, Y., Jiang, X., Sheng, S., Li, Q., & Gao, R. (2023). Normal workflow and key strategies for data cleaning in real-world research.

**Interactive Journal of Medical Research*, 12*, e44310. <https://doi.org/10.2196/44310>

R Core Team. (2025). **R: A language and environment for statistical computing**. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>

Sabo, John L., and Leah R. Gerber. "Trophic ecology." Jan. 2021, *AccessScience*, McGraw Hill, <https://doi.org/10.1036/1097-8542.711650>.

Kawakami, E., & Vazzoler, G. (1980). Método gráfico e estimativa de índice alimentar aplicado no estudo de alimentação de peixes. **Boletim do Instituto Oceanográfico*, 29*(2), 205–207. <https://www.scielo.br/j/bioce/a/dXt8qsZgcLtWDG3YgDLm9Yk/?lang=pt>

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., & Yutani, H. (2019). Welcome to the tidyverse. **Journal of Open Source Software*, 4*(43), 1686. <https://doi.org/10.21105/joss.01686>

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). **dplyr: A grammar of data manipulation** [R package version 1.1.4]. <https://CRAN.R-project.org/package=dplyr>

Wickham, H., Hester, J., & Bryan, J. (2024). **readr: Read rectangular text data** [R package version 2.1.5]. <https://CRAN.R-project.org/package=readr>